

Performance Testing

Date	30 September 2026
Team ID	SWTID-2026-3750
Project Name	ComicCraftAI
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter, [e.g. k6, Postman]
Type of Testing	[e.g., Load Testing, Stress Testing, Spike Testing]
Target Module / API	User Registration API. or endpoint was tested]
Test Environment	Cloud (AWS) [local / Hosted Server]
Test Date	

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Simulate 500 concurrent user logins	500	3600	Response time < 2s for 95%
2	Simulate 100 concurrent user logins	400	3600	Response time < 2s for 95%
3	Simulate 100 concurrent user response	500	3600	Response time < 2s for 95%
4	Simulate 100 concurrent user results	200	3600	Response time < 2s for 95%

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 seconds	Pass	Within acceptable limits
2	Response Time (Max)	< 5 seconds	3.8 seconds	Pass	Peak spikes handled well
3	Throughput (Req/sec)	≥ 100 req/sec	115 req/sec	Pass	Exceeded the performance goal
4	Error Rate	< 1%	0.2%	Pass	Very low error rate observed
5	CPU Utilization	< 80%	65%	Pass	CPU usage is stable
6	Memory Utilization	< 80%	58%	Pass	Memory consumption is optimal

Step 4: Observations & Analysis

Key Findings:

The system exhibited excellent performance, meeting all key performance indicators. The response times were well below the defined thresholds, and the throughput was consistent throughout the load test. Stability was maintained under maximum load.

Bottlenecks Identified:

Minor response time degradation was observed during the initial ramp-up, and a potential database query inefficiency was noted under heavy concurrent load.

Optimization Steps Taken:

We have optimized the heavy database query identified and adjusted the application server's connection pool settings to improve initial scaling performance.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

